COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 1.2

PROPOSITIONAL EQUIVALENCE

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TAUTOLOGY

- ***Tautology***: A compound proposition that is always true is called a tautology.

  Examples:

  a) $p \vee \neg p$

  b) The professor is either a woman or a man

  c) People either like watching TV or they don't

- Q: Show that $[\neg p \wedge (p \vee q)] \rightarrow q$ is a *tautology*.

  Solution: We can show it in **two ways**:

  (1) Using a **truth table** – show that $[\neg p \wedge (p \vee q)] \rightarrow q$ is always true

  (2) Using a **proof** (we will get to this later)

TAUTOLOGY BY TRUTH TABLE

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	F	F	T

Since the truth table shows all the true values of compound proposition $[\neg p \wedge (p \vee q)] \rightarrow q$ are true(T), so it is a tautology.

CONTRADICTION & CONTINGENCY

- ***Contradiction***: A compound proposition that is always false is called a contradiction.

- Examples:

  a) $p \wedge \neg p$

  b) x is PRIME and x is an EVEN integer greater than 8

  c) All men are good and all men are bad

- ***Contingency***: A compound proposition that is neither a tautology nor a contradiction is called a contingency.

- In other words, a compound proposition whose truth value isn't constant is called a contingency.

- Example: $p \rightarrow \neg p$

TAUTOLOGY AND CONTRADICTION

How To Determine Whether A Compound Proposition is A **Tautology** Or **Contradiction**?

- The **easiest** way to see if a compound proposition is a tautology or contradiction is to use a ***truth table***.

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TABLE 1 Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

- Compound propositions that have the same truth values in all possible cases are called **logically equivalent**.
- ***Definition*** : Compound propositions p and q are *logically equivalent* if $p \leftrightarrow q$ is a tautology (denoted by $\boldsymbol{p \equiv q}$ or $p \Leftrightarrow q$)

TABLE OF LOGICAL EQUIVALENCES

p. 17, Rosen

TABLE 5 Logical Equivalences.

Equivalence	*Name*
$p \wedge \mathbf{T} \iff p$ $p \vee \mathbf{F} \iff p$	Identity laws
$p \vee \mathbf{T} \iff \mathbf{T}$ $p \wedge \mathbf{F} \iff \mathbf{F}$	Domination laws
$p \vee p \iff p$ $p \wedge p \iff p$	Idempotent laws
$\neg(\neg p) \iff p$	Double negation law
$p \vee q \iff q \vee p$ $p \wedge q \iff q \wedge p$	Commutative laws
$(p \vee q) \vee r \iff p \vee (q \vee r)$ $(p \wedge q) \wedge r \iff p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \iff \neg p \vee \neg q$ $\neg(p \vee q) \iff \neg p \wedge \neg q$	De Morgan's laws

$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

$p \rightarrow q \equiv \neg p \vee q$

Useful Logical Equivalence(ULE)

LOGICAL EQUIVALENCES

Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

$$\begin{aligned} \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{by ULE} \\ &\equiv \neg(\neg p) \wedge \neg q && \text{by the second De Morgan law} \\ &\equiv p \wedge \neg q && \text{by the double negation law} \end{aligned}$$

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences

$$\begin{aligned} \neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the second De Morgan law} \\ &\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the first De Morgan law} \\ &\equiv \neg p \wedge (p \vee \neg q) && \text{by the double negation law} \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the second distributive law} \\ &\equiv \mathbf{F} \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv \mathbf{F} \\ &\equiv (\neg p \wedge \neg q) \vee \mathbf{F} && \text{by the commutative law for disjunction} \\ &\equiv \neg p \wedge \neg q && \text{by the identity law for F} \end{aligned}$$

Consequently $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

LOGICAL EQUIVALENCES

- Show that $[\neg p \wedge (p \vee q)] \rightarrow q$ is a **tautology** using a series of logical equivalences.

$[\neg p \wedge (p \vee q)] \rightarrow q$	
$\equiv [(\neg p \wedge p) \vee (\neg p \wedge q)] \rightarrow q$	**Distributive**
$\equiv [\, F \vee (\neg p \wedge q)] \rightarrow q$	**Negation**
$\equiv [\neg p \wedge q\,] \rightarrow q$	**Identity**
$\equiv \neg [\neg p \wedge q\,] \vee q$	**ULE**
$\equiv [\neg(\neg p) \vee \neg q\,] \vee q$	**De Morgan**
$\equiv (p \vee \neg q\,) \vee q$	**Double Negation**
$\equiv p \vee [\neg q \vee q\,]$	**Associative**
$\equiv p \vee T$	**Domination**
$\equiv T$	

So, $[\neg p \wedge (p \vee q)] \rightarrow q$ is a tautology

LOGICAL EQUIVALENCES

Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$(p \wedge q) \rightarrow (p \vee q)$	$\equiv \neg(p \wedge q) \vee (p \vee q)$	by Example 3
	$\equiv (\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
	$\equiv (\neg p \vee p) \vee (\neg q \vee q)$	by the associative and commutative laws for disjunction
	$\equiv \mathbf{T} \vee \mathbf{T}$	by Example 1 and the commutative law for disjunction
	$\equiv \mathbf{T}$	by the domination law

REFERENCES

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- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley